Expt 6	Finite Word length Effects
Date:	6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

```
clc;
clear;
close all;

%% Signal
n = 0:79;
x = sin(2*pi*n/40);

%% Quantization
q = 8;

% Truncation
xt = floor(x*q)/q;

% Rounding
xr = round(x*q)/q;

%% Quantization Error
et = x - xt;
er = x - xr;

%% Figure 1: Original vs Quantized
figure;
plot(n, x, 'b', n, xr, 'r', 'LineWidth', 1.5);
```

```
grid on;
title('Original vs Quantized');
xlabel('Sample Number');
ylabel('Amplitude');
legend('Original', 'Quantized');

%% Figure 2: Truncation vs Rounding
figure;
plot(n, x, 'b', n, xt, 'r', n, xr, 'g', 'LineWidth', 1.5);
grid on;
title('Truncation vs Rounding');
xlabel('Sample Number');
ylabel('Amplitude');
legend('Original', 'Truncated', 'Rounded');

%% Figure 3: Quantization Error
figure;
stem(n, et, 'r', 'filled');
hold on;
stem(n, er, 'g', 'filled');
grid on;
title('Quantization Error');
xlabel('Sample Number');
ylabel('Error');
legend('Truncation', 'Rounding');
hold off;

%% Figure 4: MSE Comparison
```

```
MSE_truncation = mean(et.^2);
MSE_rounding = mean(er.^2);

figure;
bar([MSE_truncation MSE_rounding]);

set(gca, 'XTick', 1:2);
set(gca, 'XTickLabel', {'Truncation', 'Rounding'});

grid on;
title('MSE Comparison');
ylabel('MSE');
```

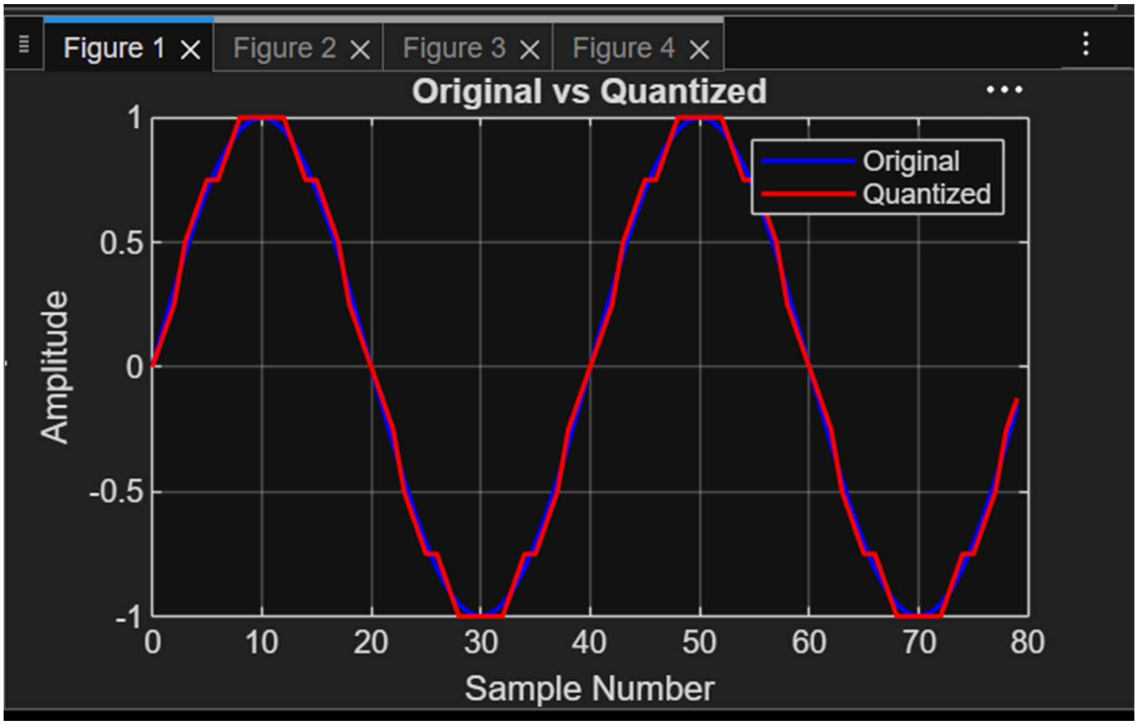
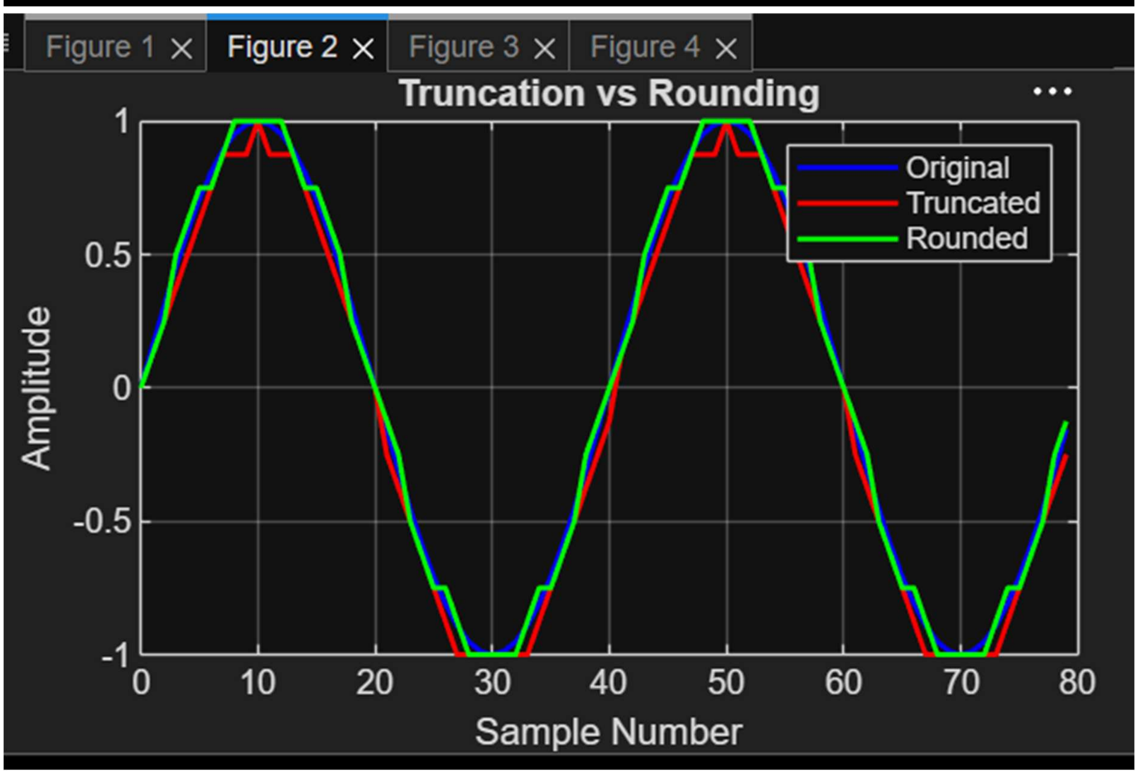


Figure 1
Figure 2
Figure 3
Figure 4
Original vs Quantized
Original
Quantized
Amplitude
Sample Number
1
0.5
0
-0.5
-1
0
10
20
30
40
50
60
70
80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Original
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Amplitude
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1
0.5
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-0.5
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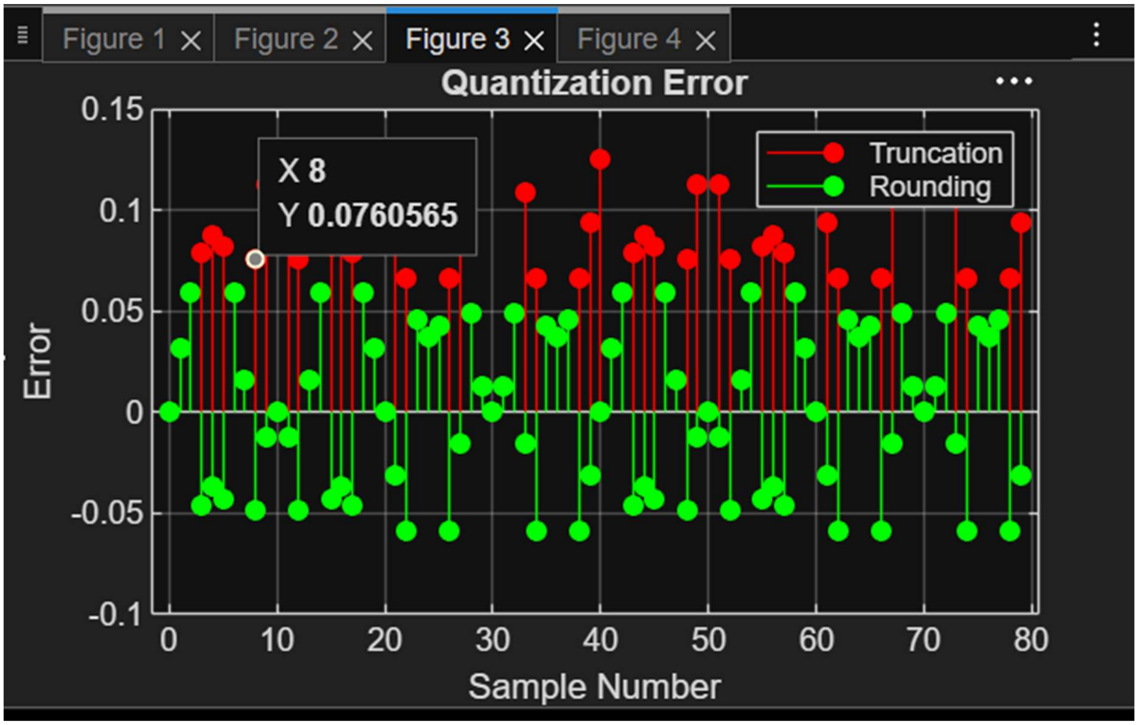

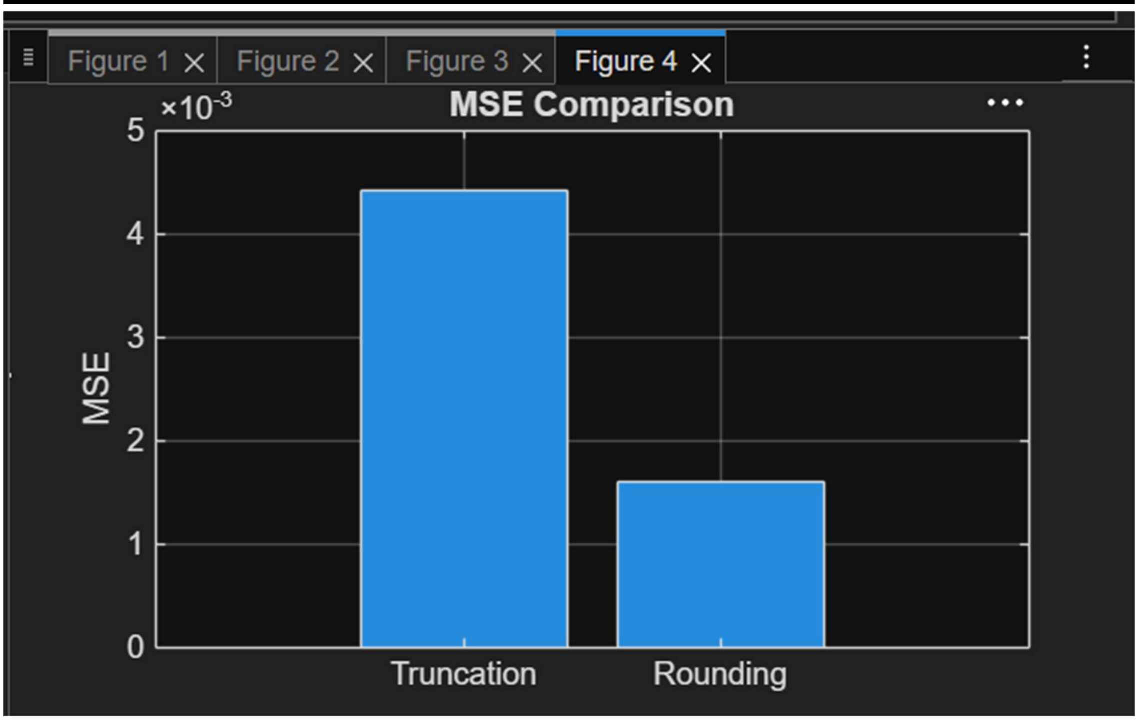

Result: 1. The digital signal was successfully represented using a finite word length format in MATLAB.

2. Truncation and rounding operations were implemented on the signal samples.

3. The errors introduced by truncation and rounding were computed and compared, showing that rounding generally produces smaller errors than truncation.